Machine Learning Frontiers

Review of what we've covered

Challenges and open questions

Interesting applications

Where to go from here?

Supervised Learning
K-Nearest Neighbors
Linear regression
Perceptron
Logistic regression
Discriminant Analysis
Naïve Bayes
Classification and Regression Trees
Random Forests
Neural Networks, backpropagation
Deep Learning methods
    Convolutional neural networks
    Transformers

Ensemble methods
    Bagging, boosting, stacking
Regularization (ridge and lasso), feature selection, cost functions, and norms
Decision theory
Gradient descent and stochastic gradient descent

Performance Evaluation
Cross validation
Bootstrap sampling
Confusion Matrices
ROC curves
Precision/Recall/Error Types
Bias-variance tradeoff
Curse of Dimensionality

Unsupervised Learning
Clustering
    K-Means
    Gaussian mixture model
    Agglomerative clustering
    DBSCAN
    Spectral clustering
Density Estimation
    GMMs, Kernel Density Estimation, Histograms
Dimensionality Reduction
    PCA, UMAP, t-SNE

Markov Models
Markov chains
Markov reward processes
Markov Decision Processes

Reinforcement Learning
Dynamic Programming
Policy Evaluation
Policy Improvement
Policy Iteration
Value Iteration
Generalized policy iteration
Monte Carlo Control

Semi-supervises learning
Transfer learning
Self-supervised learning

Topics we covered

Machine Learning is NOT magic

It's **one tool of many** to assist us in making better, faster decisions

It is not a substitute for critical thinking and domain expertise – they amplify each other

Practical Considerations for Machine Learning

1. Let your **problem/question** drive your design choices
2. Set a reasonable goal and clear metric of success
3. Ask yourself if there are **non-ML approaches that would work?**
4. Develop an **end-to-end pipeline** as soon as you're able and keep it maintained (data preparation & preprocessing, analysis, and performance evaluation)
5. Start with the **simplest solution** you can and layer on complexity as needed

- **Features / representations** are often more important than algorithms
- **Experimental design** is often more important than algorithms

Adapted from Google: https://developers.google.com/machine-learning/guides/rules-of-ml

More advice

- ALWAYS look at your data – before you begin, the inputs/outputs, etc.
  - Check your distributions
  - Explore outliers to get insights on the model
- Report confidence intervals whenever possible (make sure your "better" model is not just a noisy aberration)
- When comparing supervised models, make sure your comparing on the same validation set
- Make sure you NEVER mix training and validation information

Adapted from https://www.unofficialgoogledatascience.com/2016/10/practical-advice-for-analysis-of-large.html

Model Selection Workflow: Data Type

Tabular	Imagery	Text
Gradient boosting (XGBoost, LightGBM, CatBoost)	Convolutional Neural Networks (CNNs) (ResNet or EfficientNet variants)	Transformers (BERT variants like DeBERTa)
Random forests	Vision Transformers (ViTs) (SWIN Transformer or DINOv2/v3 as a universal backbone)	
Linear models (often as baselines for comparison or when interpretability is needed)		

Note: generative text/image models are not included here. This assumes we want deterministic output

Always start with default hyperparameters and ONLY tune if needed

Challenges and Open Questions

Unsolved challenges in machine learning

Generalizing from **small numbers of examples** (few-shot learning)

Adapting to **new environments** and non-stationary problems

Transferring knowledge between tasks (transfer learning)

Combining **heterogeneous data sources** into a model

Interpretability for confidence in algorithms

Ethics, fairness, and privacy (asks the question: should we?)

Greenwald and Oertel, 2017, Future Directions in Machine Learning
Jeannette Wing, 2020, Ten Research Challenge Areas in Data Science

Types of Biases

- **Representation bias**
- **Sampling bias**
- Historical bias
- Measurement bias
- Evaluation bias
- Aggregation bias
- Population bias
- Simpson's paradox
- Behavioral bias
- Longitudinal data fallacy
- Content production bias
- Linking bias
- Temporal bias
- Popularity bias
- Algorithmic bias
- User interaction bias
- Presentation bias
- Ranking bias
- Social bias
- Emergent bias
- Self-selection bias
- Omitted variable bias
- Cause-effect bias
- Observer bias
- Funding bias

Bold indicates those biases we've discussed through the semester – there are MANY more

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A., 2019. A survey on bias and fairness in machine learning. arXiv preprint arXiv:1908.09635.

Interpretability

Transparency
(can I tell how the model works)

- **Simulatability**: can I contemplate the whole model at once?
- **Decomposability**: is there an intuitive explanation for each part of the model?
  (e.g. all patients with diastolic blood pressure over 150)

Explainability
(explaining outputs, not necessarily understanding models)

Visualization, local explanations, explanations by example

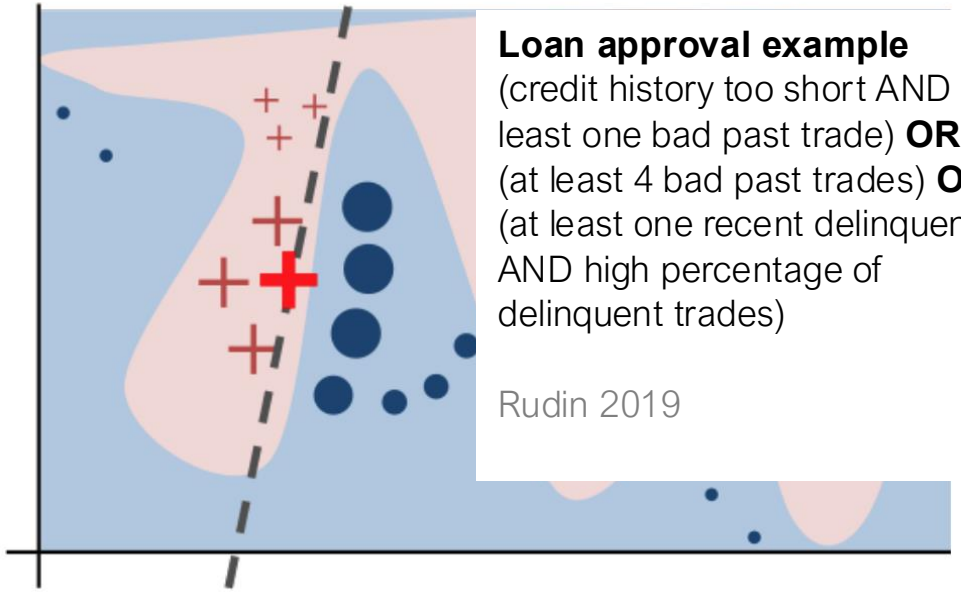
(e.g. this tumor is classified as malignant because to the model it looks a lot like these other tumors)

Lipton, Zachary C. "The Mythos of Model Interpretability: In Machine Learning, the Concept of Interpretability Is Both Important and Slippery." Queue 16, no. 3 (2018): 31–57.

Recidivism prediction algorithm
Performance as good as a black box model with 130+ factors; might include socio-economic info; expensive (software license); within software used in US justice system

IF	age between 18–20 and sex is male	THEN predict arrest (within 2 years)
ELSE IF	age between 21–23 and 2–3 prior offences	THEN predict arrest
ELSE IF	more than three priors	THEN predict arrest
ELSE	predict no arrest	

Rudin, Cynthia. "Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead." Nature Machine Intelligence 1, no. 5 (2019): 206–15.



Loan approval example
(credit history too short AND at least one bad past trade) **OR** (at least 4 bad past trades) **OR** (at least one recent delinquency AND high percentage of delinquent trades)

Rudin 2019

Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "Model-Agnostic Interpretability of Machine Learning." ArXiv Preprint ArXiv:1606.05386, 2016.

Model Selection Workflow: Interpretability

Explain any model

SHAP
SHapley Additive exPlanations

LIME
Local Interpretable Model-agnostic Explanations

Interpretable / transparent models

Linear models
Interpretable for simple linear relationships

Decision Trees (sparse)
Good for simple nonlinear relationships in an intuitive. For more performant decision trees, use Sparse Optimal Decision Trees

Generalized Additive Models (GAMs)
A balanced option between the most interpretable models and black box models

Explainable Boosting Machines
Learns GAMs with less manual tuning

AI Alignment

"Artificial intelligence (AI) alignment is the process of **encoding human values and goals** into AI models to make them as **helpful, safe and reliable** as possible"
(IBM)

Facebook in Myanmar

Facebook's algorithmic goal: maximize user engagement

Posts that generated outrage, fear, and sensationalism also generated engagement and were amplified

Also, there was a lack of Burmese language/context understanding which made automated content moderation difficult

The New York Times

Facebook Admits It Was Used to Incite Violence in Myanmar

Share full article



Rohingya refugees after crossing the Naf River, which separates Myanmar and Bangladesh, in 2017. A report commissioned by Facebook found the company failed to keep its platform from being used to "foment division and incite offline violence" in Myanmar. Adam Dean for The New York Times

https://www.nytimes.com/2018/11/06/technology/myanmar-facebook.html

Ethical Considerations

- Privacy and surveillance
- Manipulation of behavior
- Transparency in machine learning systems
- Decision system biases
- Automation and employment
- Autonomous systems and responsibility
- Machine ethics (e.g. Isaac Asimov's 3 laws)
- Artificial moral agents
- Singularity

Müller, V.C., 2020. Ethics of artificial intelligence and robotics.

Computational Challenges

NVIDIA
Graphics Processing Units (GPU)



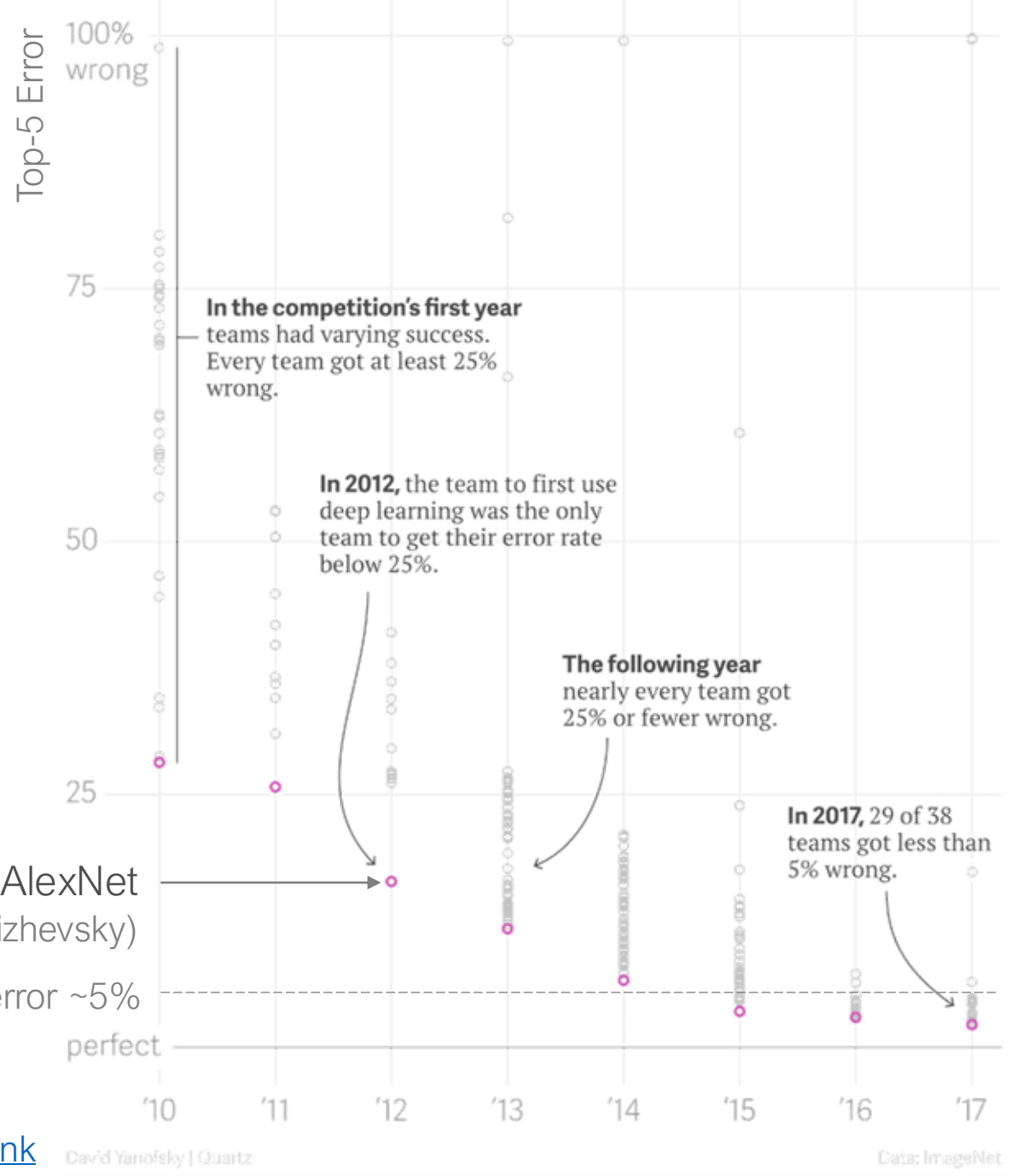
Google
Tensor Processing Unit (TPU)



ImageNet Large Scale Visual Recognition Challenge (ILSVRC)

Fei-Fei Li et al. 2010 (link)

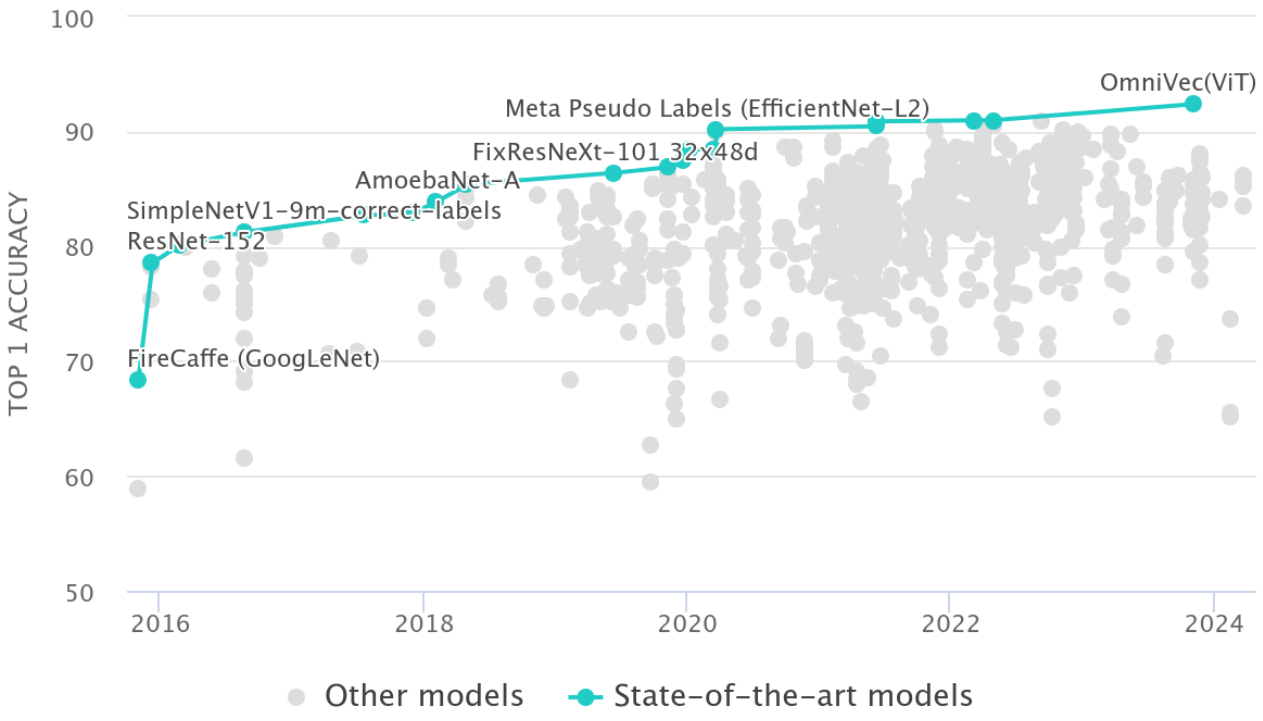
Competition at:
Conference on Computer Vision and Pattern Recognition (CVPR)

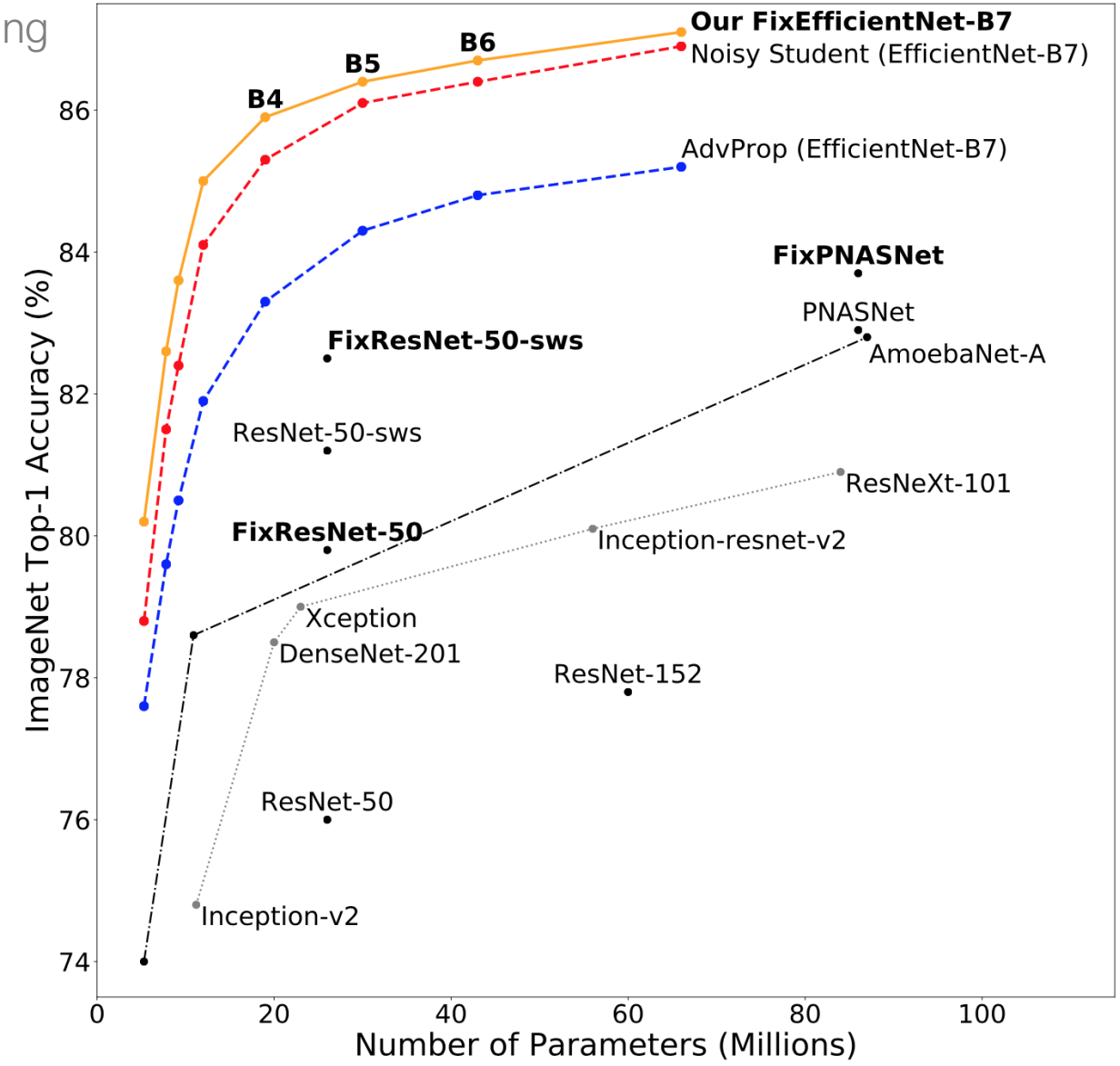


Source: Quartz, link

Latest Advances with Image Recognition

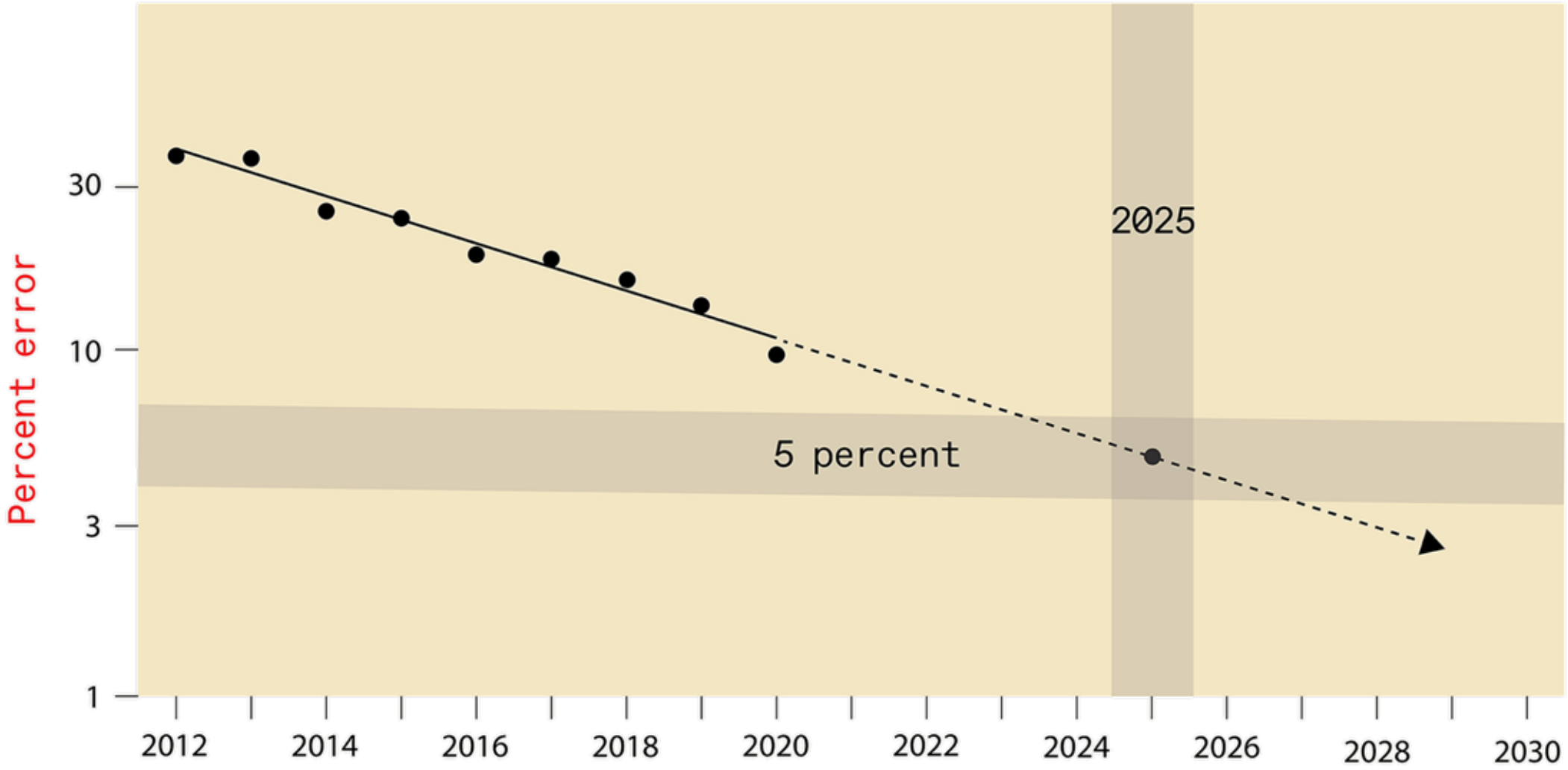
Touvron, H., Vedaldi, A., Douze, M. and Jégou, H., 2020. Fixing the train-test resolution discrepancy: FixEfficientNet. arXiv preprint arXiv:2003.08237.





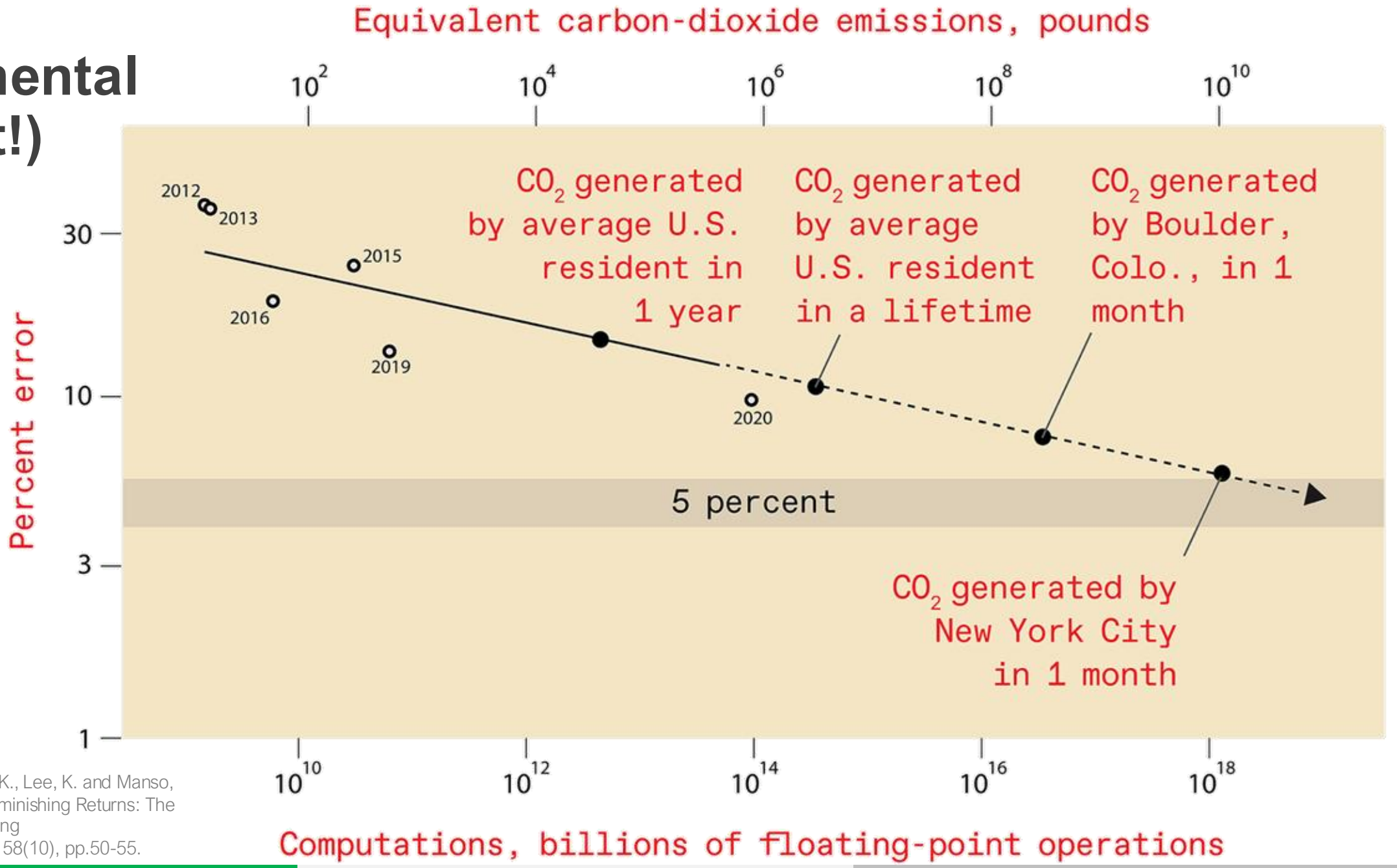
Source: https://paperswithcode.com/sota/image-classification-on-imagenet

Trend in ImageNet performance



Thompson, N.C., Greenewald, K., Lee, K. and Manso, G.F., 2021. Deep Learning's Diminishing Returns: The Cost of Improvement is Becoming Unsustainable. IEEE Spectrum, 58(10), pp.50-55.

Potential environmental (and cost!) impacts



Thompson, N.C., Greenewald, K., Lee, K. and Manso, G.F., 2021. Deep Learning's Diminishing Returns: The Cost of Improvement is Becoming Unsustainable. IEEE Spectrum, 58(10), pp.50-55.

The Bitter Lesson (link,video)

Richard Sutton

"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin."

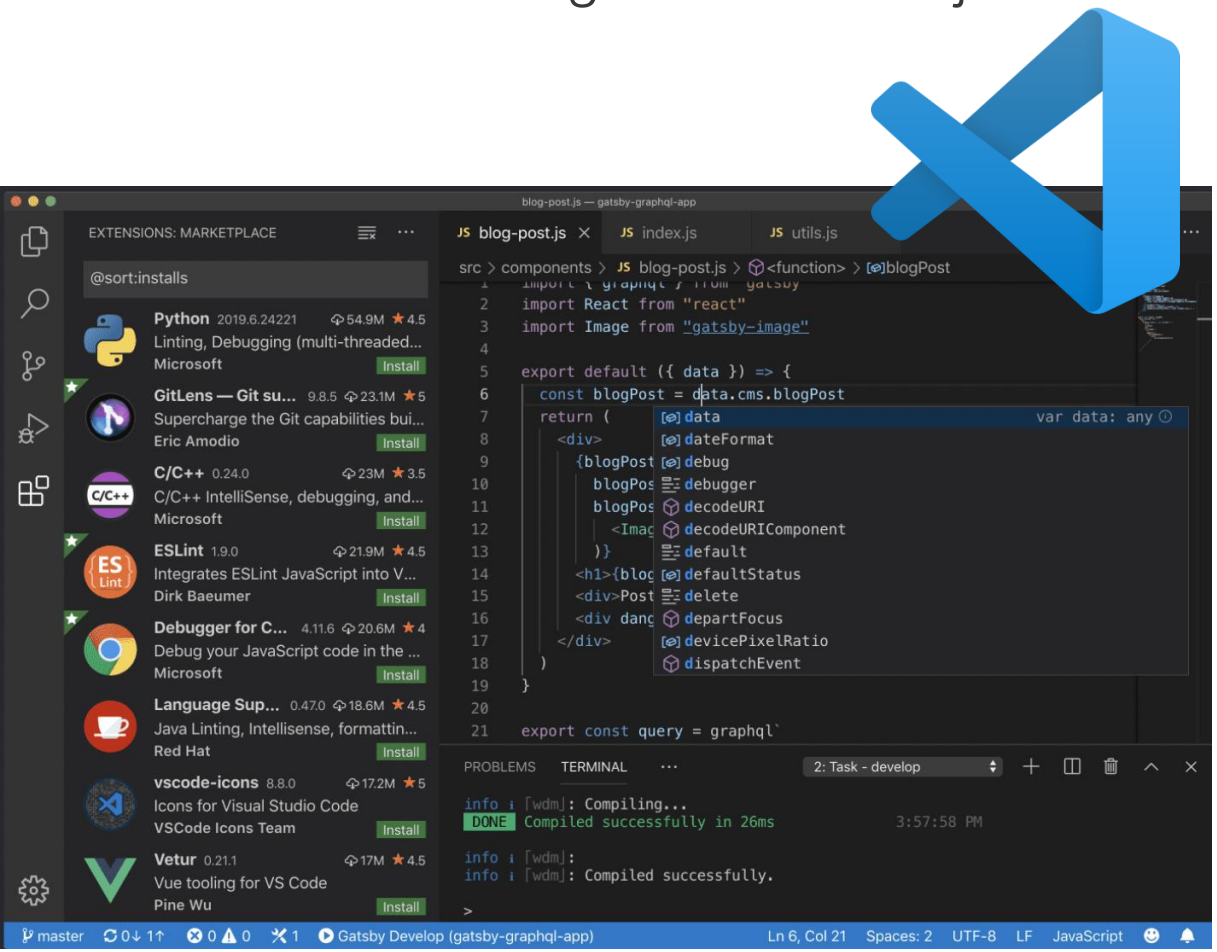
How to deal with increasing complexity?

More code

More machine learning experiments to keep track of

Tools: Development Environments

Use the best coding tools for the job: VS Code, Jupyter, etc.





For additional practical considerations, explore Berkeley's Full Stack Deep Learning

Organization: Experiment Tracking

Track your experiments in an organized way. If it helps, use Weights and Biases, Comet.ml, MLFlow, Neptune, etc.

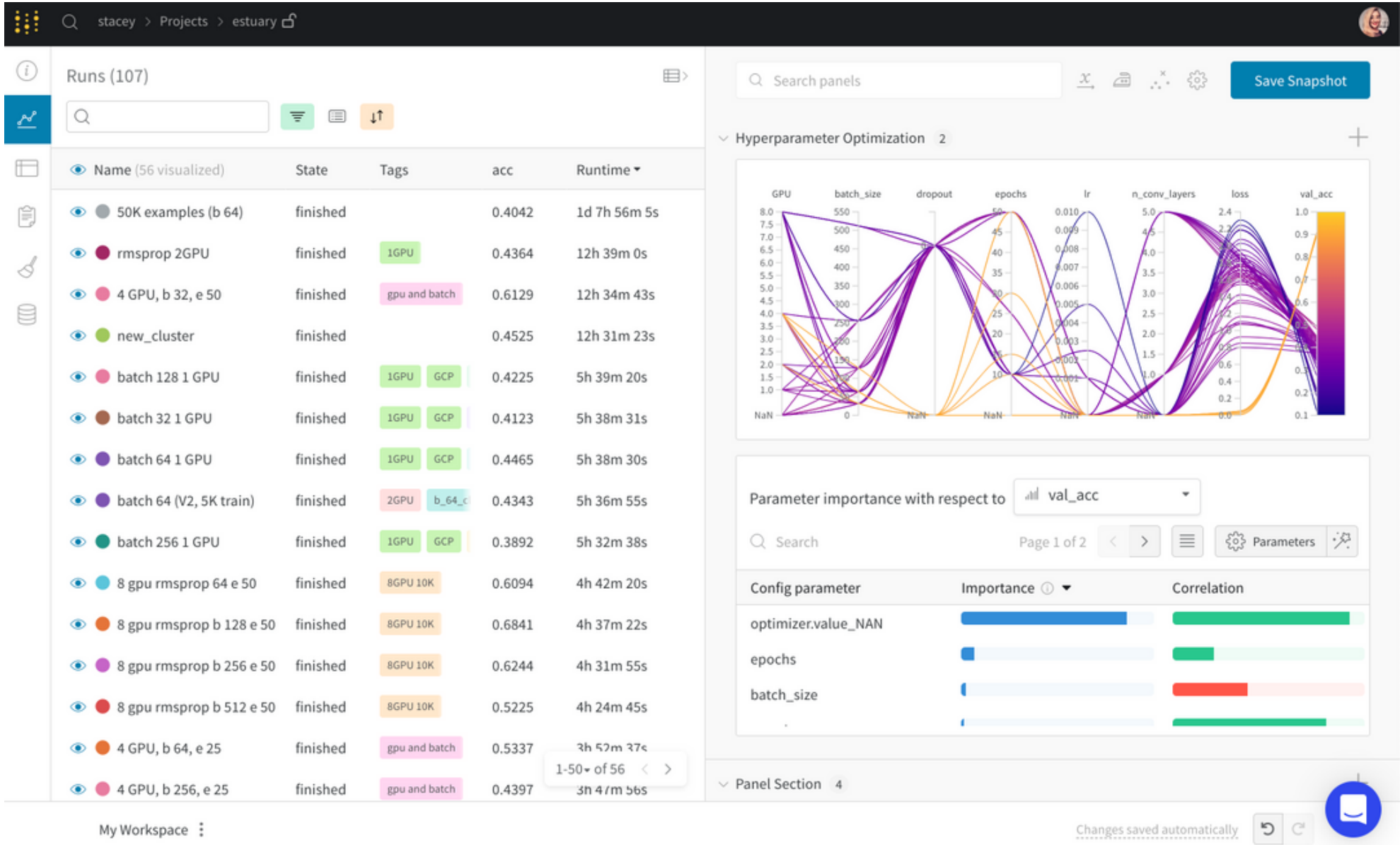


Image from:
https://github.com/wandb/client

File Organization

Cookiecutter Data Science
https://cookiecutter-data-science.drivendata.org/

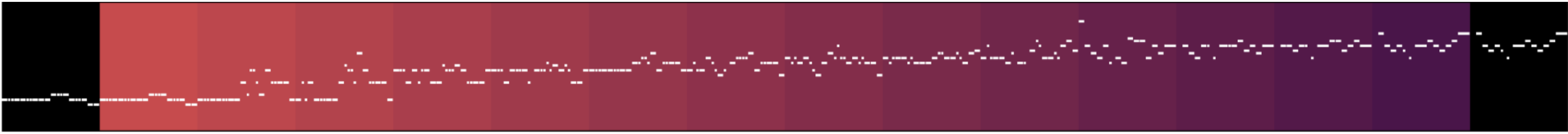
```
├── LICENSE            <- Open-source license if one is chosen
├── Makefile           <- Makefile with convenience commands like `make data` or `make train`
├── README.md          <- The top-level README for developers using this project.
├── data
│   ├── external       <- Data from third party sources.
│   ├── interim        <- Intermediate data that has been transformed.
│   ├── processed      <- The final, canonical data sets for modeling.
│   └── raw            <- The original, immutable data dump.
│
├── docs               <- A default mkdocs project; see www.mkdocs.org for details
│
├── models             <- Trained and serialized models, model predictions, or model summaries
│
├── notebooks          <- Jupyter notebooks. Naming convention is a number (for ordering),
│                         the creator's initials, and a short `-` delimited description, e.g.
│                         `1.0-jqp-initial-data-exploration`.
│
├── pyproject.toml     <- Project configuration file with package metadata for
│                         {{ cookiecutter.module_name }} and configuration for tools like black
│
├── references         <- Data dictionaries, manuals, and all other explanatory materials.
│
├── reports            <- Generated analysis as HTML, PDF, LaTeX, etc.
│   └── figures        <- Generated graphics and figures to be used in reporting
│
├── requirements.txt   <- The requirements file for reproducing the analysis environment, e.g.
│                         generated with `pip freeze > requirements.txt`
│
├── setup.cfg          <- Configuration file for flake8
│
└── {{ cookiecutter.module_name }}   <- Source code for use in this project.
    │
    ├── __init__.py             <- Makes {{ cookiecutter.module_name }} a Python module
    │
    ├── config.py               <- Store useful variables and configuration
    │
    ├── dataset.py              <- Scripts to download or generate data
    │
    ├── features.py             <- Code to create features for modeling
    │
    ├── modeling
    │   ├── __init__.py
    │   ├── predict.py          <- Code to run model inference with trained models
    │   └── train.py            <- Code to train models
    │
    └── plots.py                <- Code to create visualizations
```

Machine Learning Applications

Music

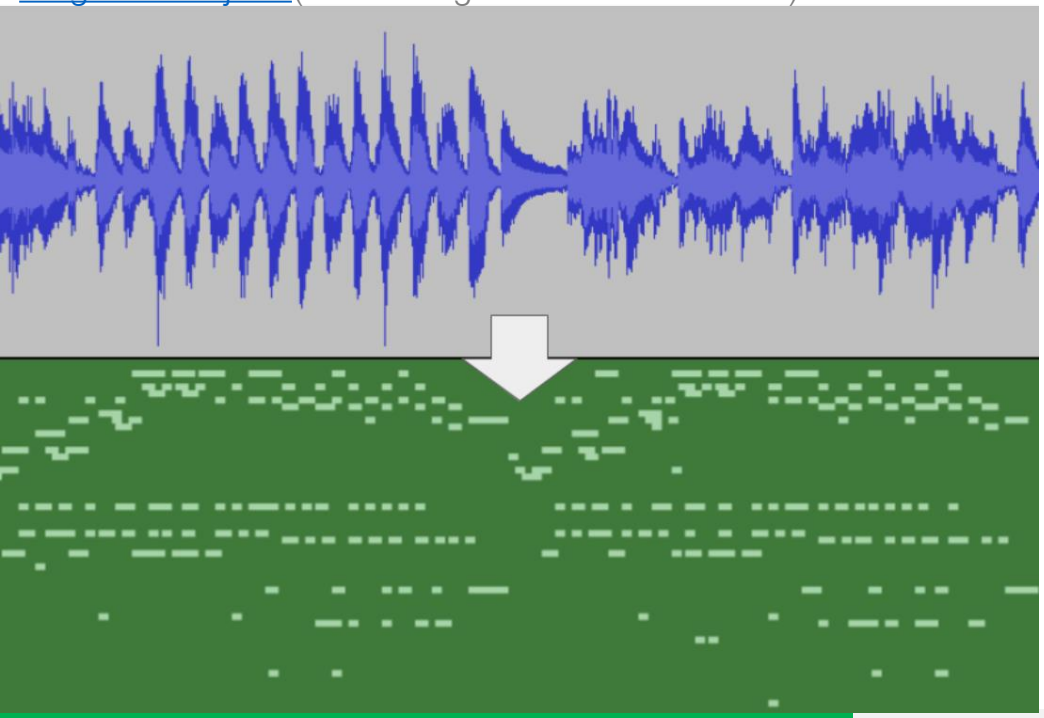
MusicVAE (Magenta):
Blending musical scores (link)

Magenta Project (from Google Brain / tensorflow)



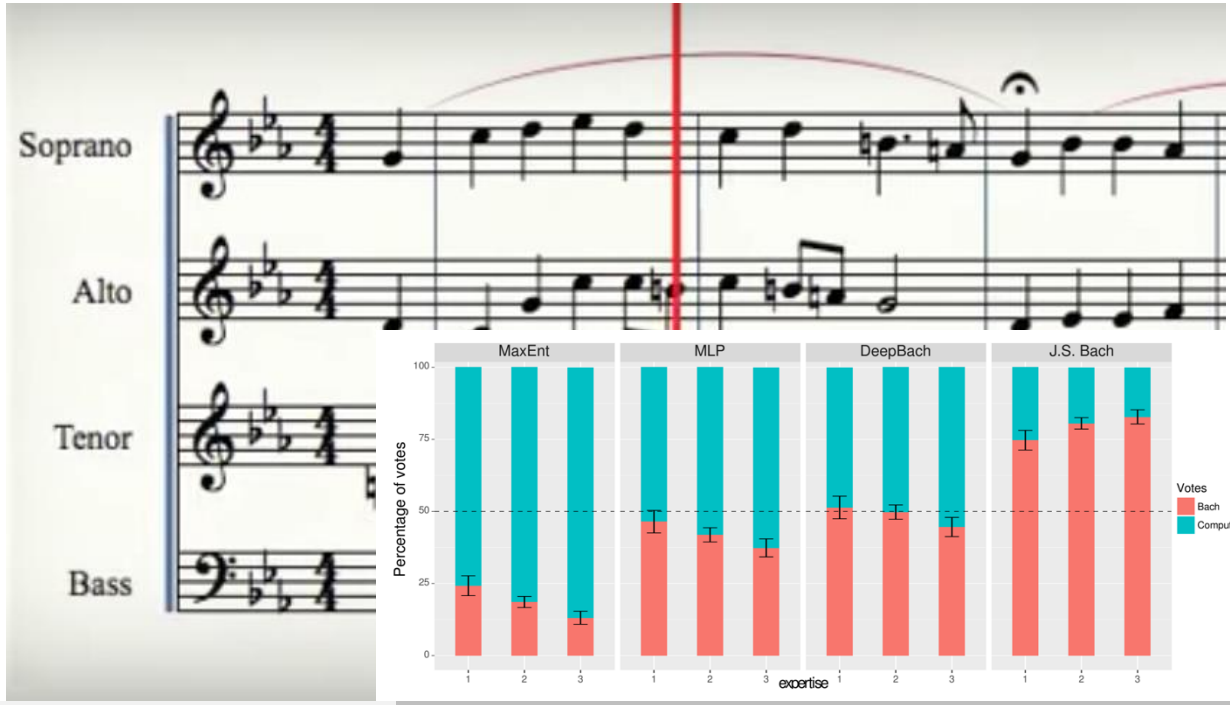
Onsets and Frames:
Automated transcription (link)

Magenta Project (from Google Brain / tensorflow)



Deep Bach:
Automated harmonization (link)

Paper available here: https://arxiv.org/abs/1612.01010



MelNet (link)

Paper available here: https://arxiv.org/abs/1906.01083

A Generative Model for Audio in the Frequency Domain *June 5, 2019*

Existing generative models for audio have predominantly aimed to directly model time-domain waveforms. MelNet instead aims to model the frequency content of an audio signal. MelNet can be used to model audio unconditionally, making it capable of tasks such as music generation. It can also be conditioned on text and speaker, making it applicable to tasks such as text-to-speech and voice conversion. The full paper is available on arXiv: https://arxiv.org/abs/1906.01083.

Demo: Select a speaker and a sentence to view a spectrogram generated by MelNet. Play the audio to visualize how the frequencies change over time.



Generation

https://thisxdoesnotexist.com/

https://thispersondoesnotexist.com/



Computer Vision & Visual Arts

Deep Dream: Style Transfer and Abstract Art (link)

Originally developed by Alexander Mordvintsev from Google

There are now websites for DIY deep art: https://deepart.io/



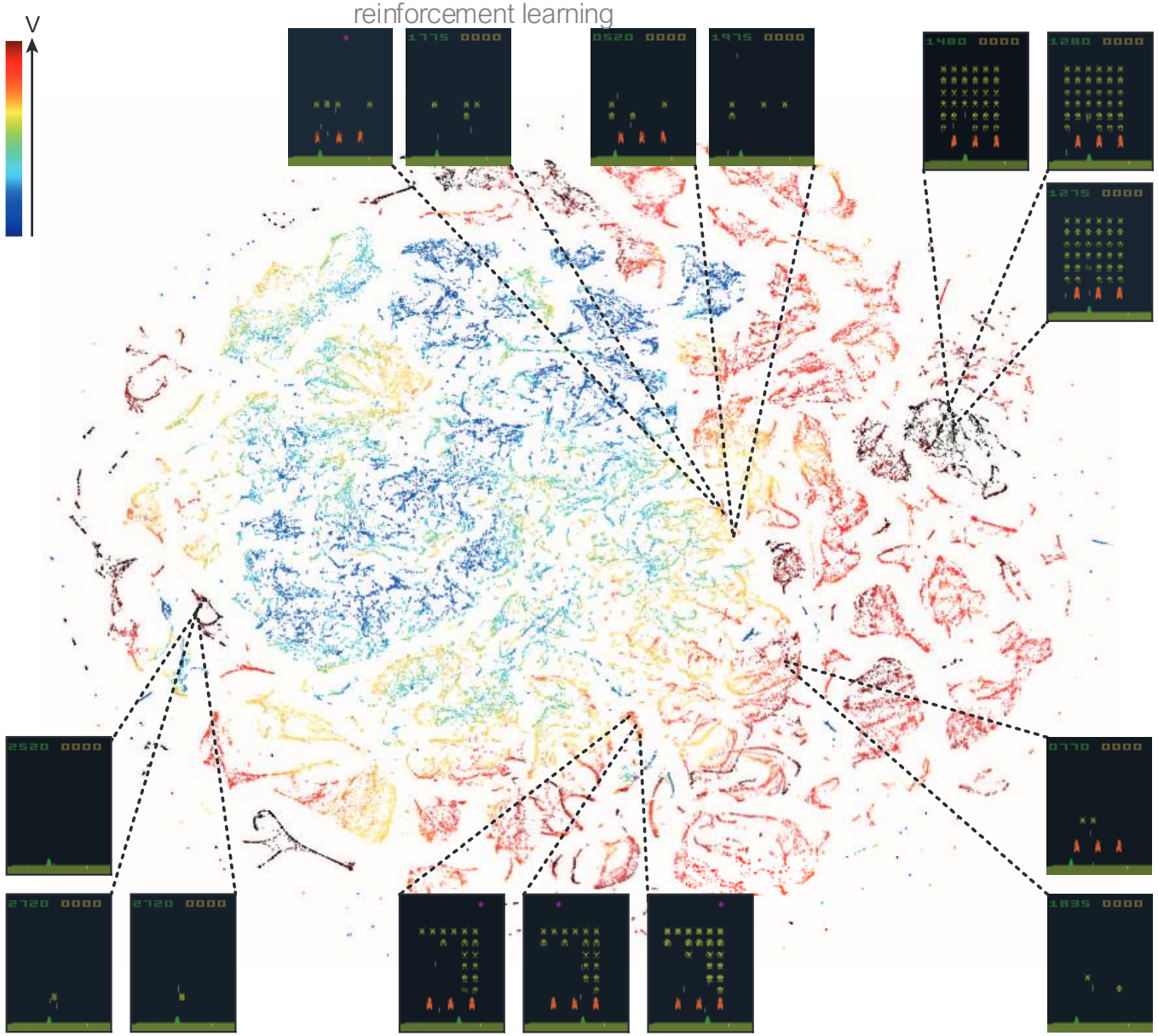
This image is from a now-defunct website https://deepart.io/ where you could purchase AI artwork:

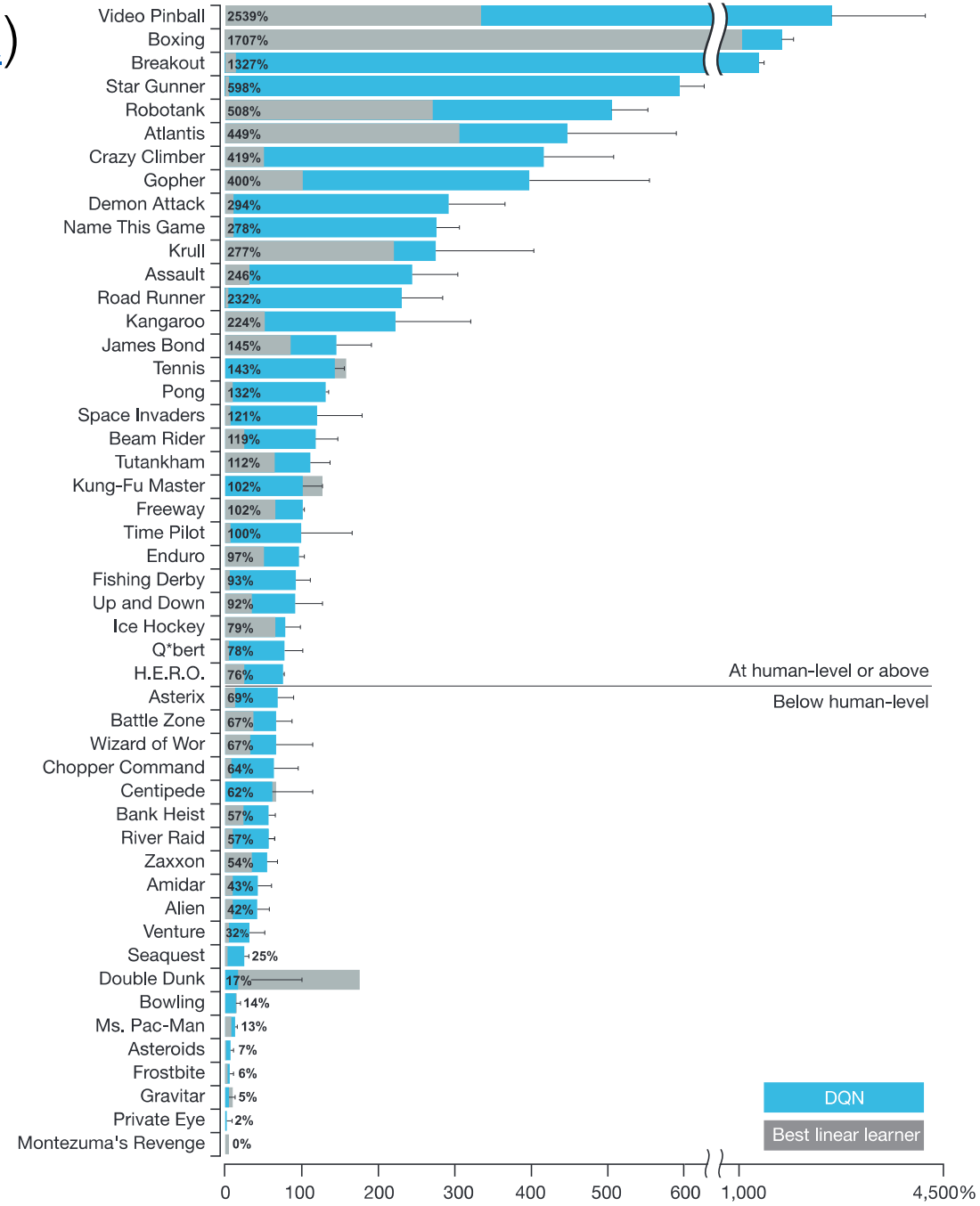


Games

Reinforcement Learning for Atari (link)

Mnih et al. 2015 (DeepMind), Human-level control through deep reinforcement learning





Games & Reinforcement Learning

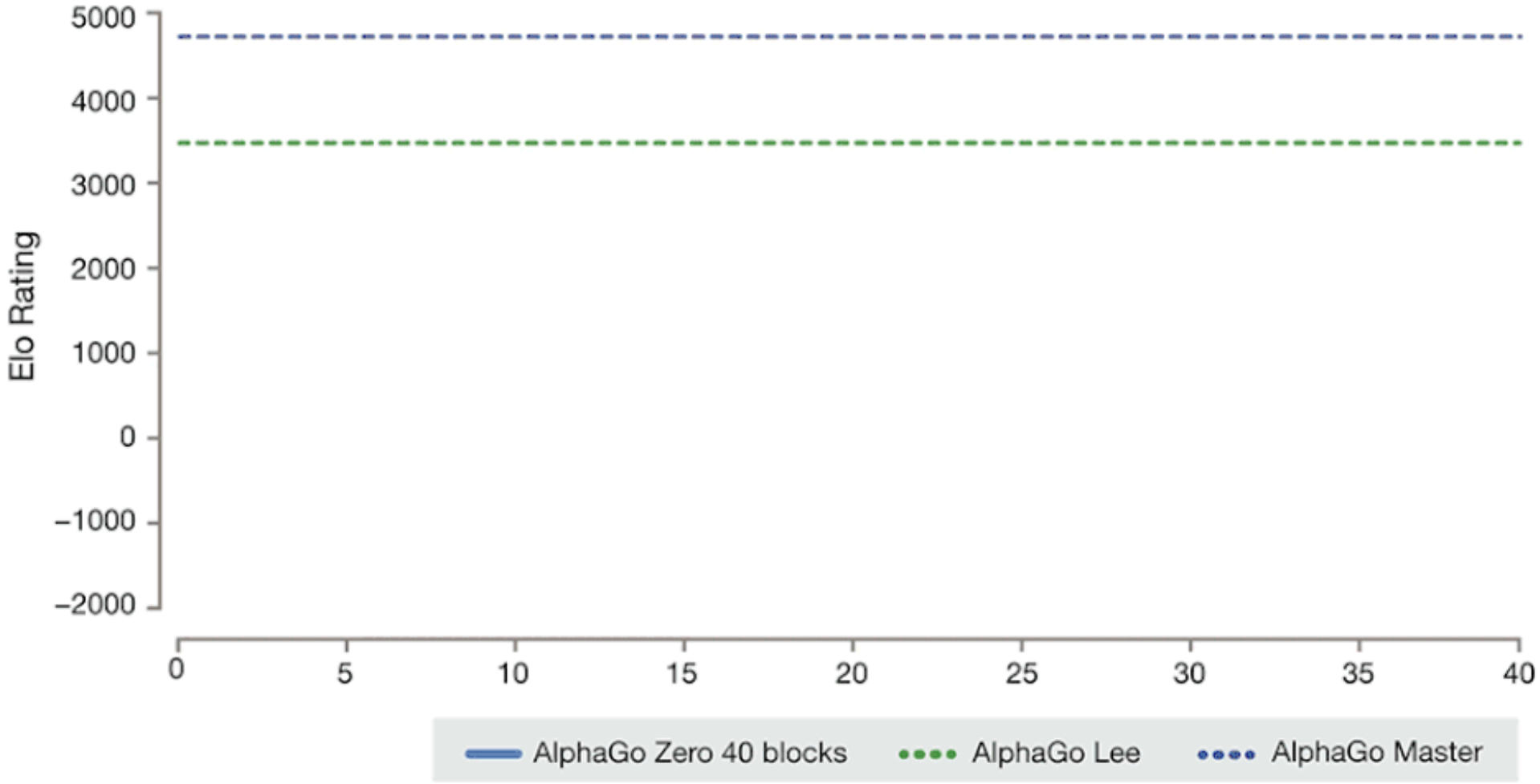
AlphaZero (link)

Silver et al. 2017 (DeepMind), Mastering Chess and Shogi by Self-Play with a General Reinforcement Learning Algorithm

Learning Go starting from random play

Mastered in 24 hours

Did the same with Chess



AlphaStar: Learning to Play StarCraft II

Vinyals, O., Babuschkin, I., Czarnecki, W.M., Mathieu, M., Dudzik, A., Chung, J., Choi, D.H., Powell, R., Ewalds, T., Georgiev, P. and Oh, J., 2019. Grandmaster level in StarCraft II using multi-agent reinforcement learning. Nature, 575(7782), pp.350-354.

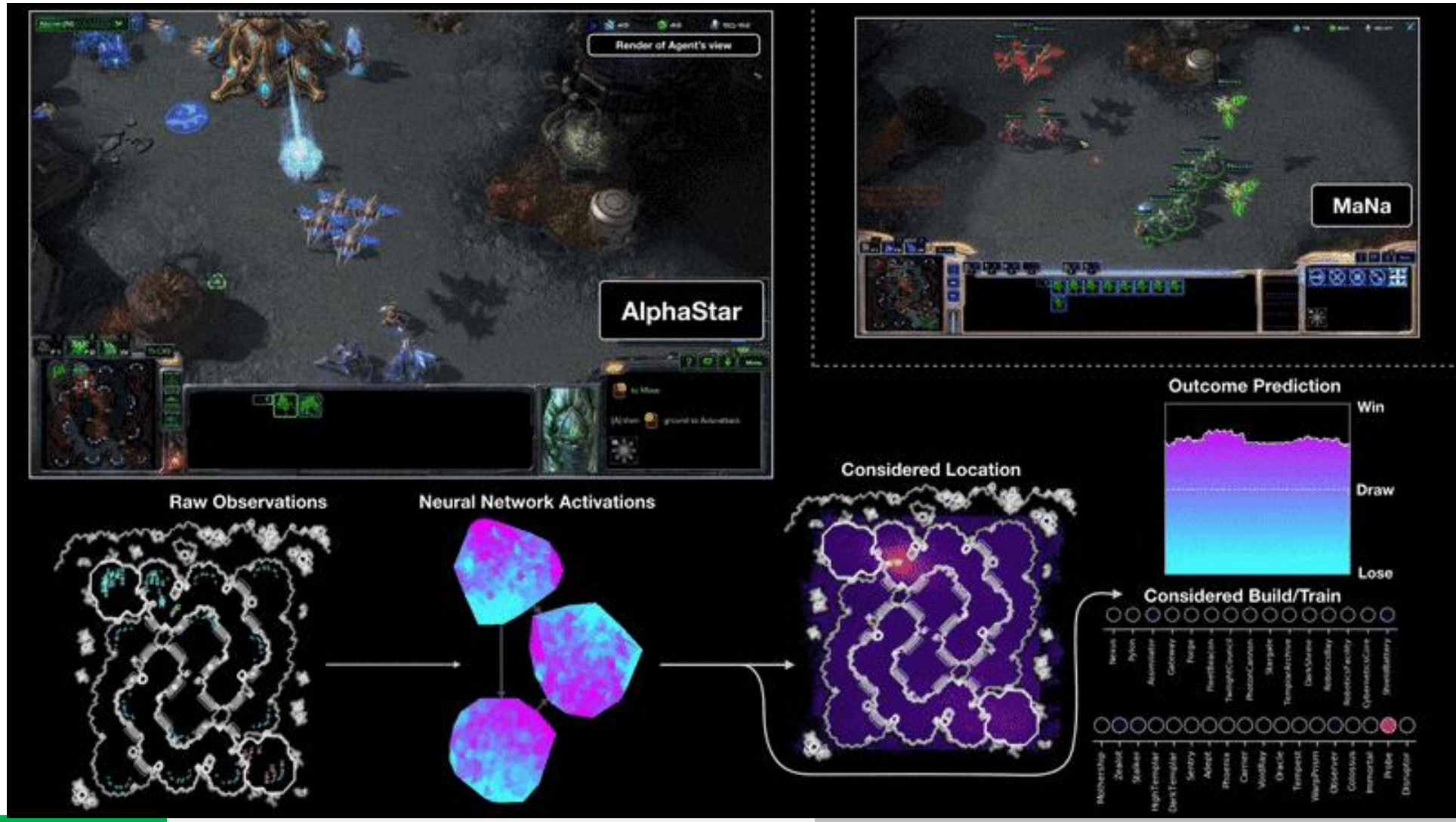


Image from https://deepmind.com/blog/article/alphastar-mastering-real-time-strategy-game-starcraft-ii

AlphaStar: Learning to Play StarCraft II

Vinyals, O., Babuschkin, I., Czarnecki, W.M., Mathieu, M., Dudzik, A., Chung, J., Choi, D.H., Powell, R., Ewalds, T., Georgiev, P. and Oh, J., 2019. Grandmaster level in StarCraft II using multi-agent reinforcement learning. Nature, 575(7782), pp.350-354.

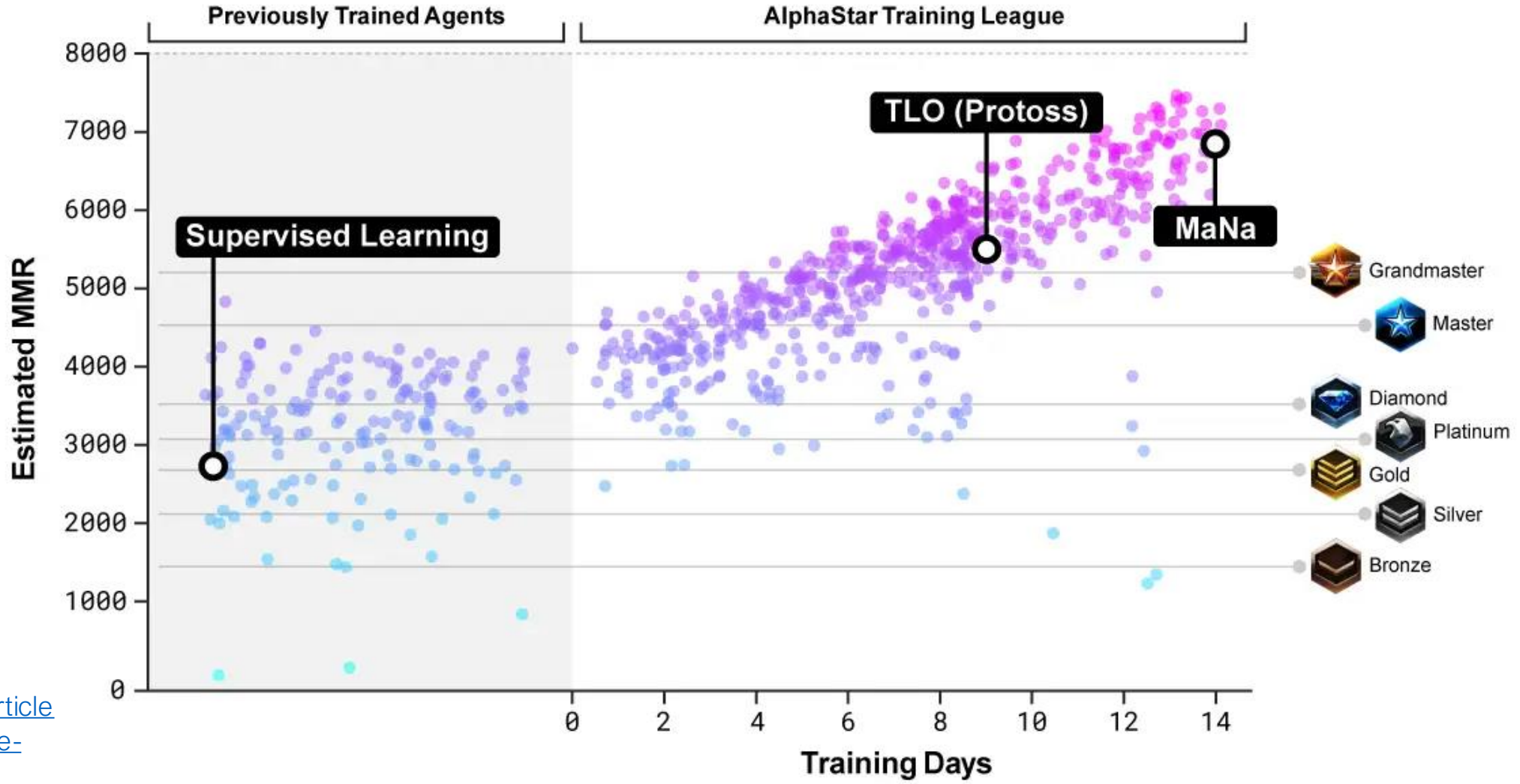


Image from https://deepmind.com/blog/article/alphastar-mastering-real-time-strategy-game-starcraft-ii

Open source frameworks

Keras (link)
Wrapper for PyTorch and Tensorflow making coding easier

PyTorch (link)
Framework for implementing graphical models, such as neural networks

Gymnasium (link)
Toolkit for developing and comparing reinforcement learning algorithms



Where to go from here?

Courses at Duke

ECE 585: Signal Detection and Extraction Theory
ECE 586: Vector Space Methods with Applications
ECE 588: Image & Video Processing
ECE 661: Computer Engineering Machine Learning and Deep Neural Nets
ECE 684: Natural Language Processing
ECE 685D / CS675D: Intro to Deep Learning
ECE 687D / CS 671D / STA 671D: Theory and Algorithms for Machine Learning
ECE 689/ CS676: Advanced Topics in Deep Learning
AIPI 531 - Reinforcement Learning
CompSci 527: Computer Vision
Math 412: Topological Data Analysis
Math 465/CompSci 445: Introduction to High Dimensional Data Analysis
Math 766: Mathematics of Machine Learning
STA 601: Bayesian and Modern Statistical Methods
STA 623: Statistical Decision Theory
STA 561D: Probabilistic Machine Learning
STA 571: Advanced Probabilistic Machine Learning
ECE 590: Special Topics courses on machine learning and artificial intelligence topics

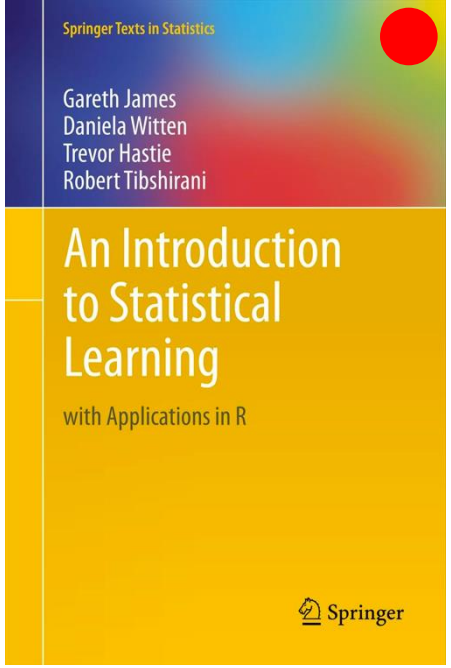
Courses on foundational concepts: Probability, Statistics, Linear Algebra, Mathematics, Programming and Software Development

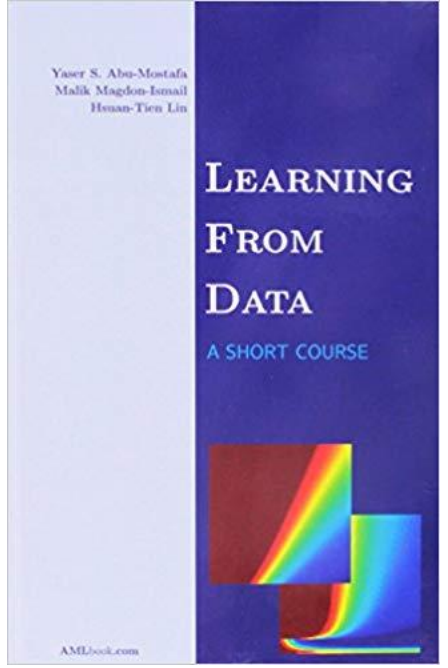
Data Science Books

Additional Resources Available at:
https://kylebradbury.org/datascience.html

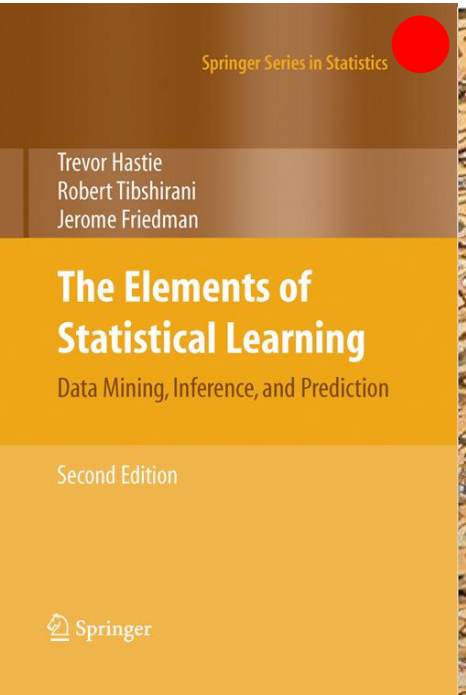
Introductory Texts

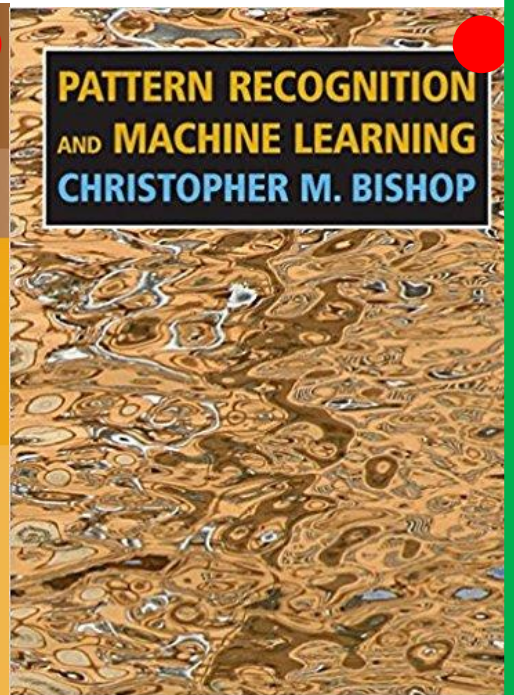
Free online

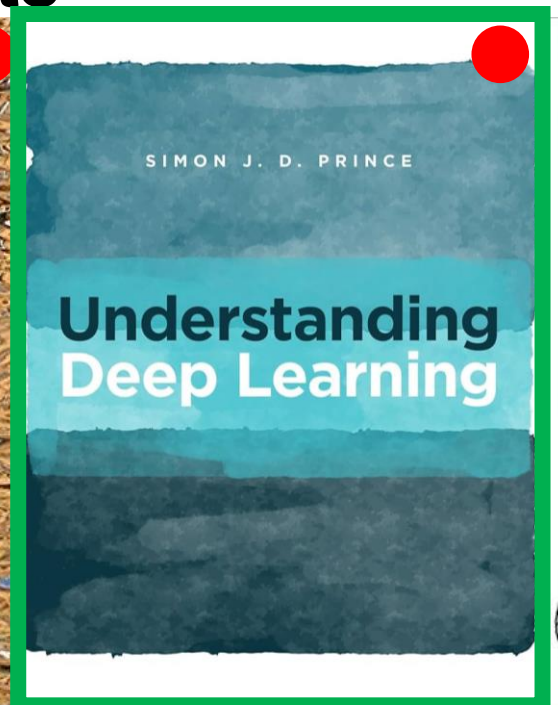




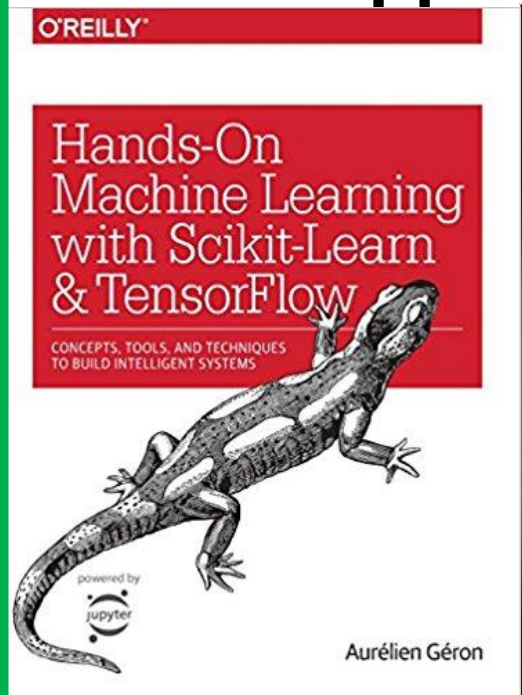
Advanced Texts

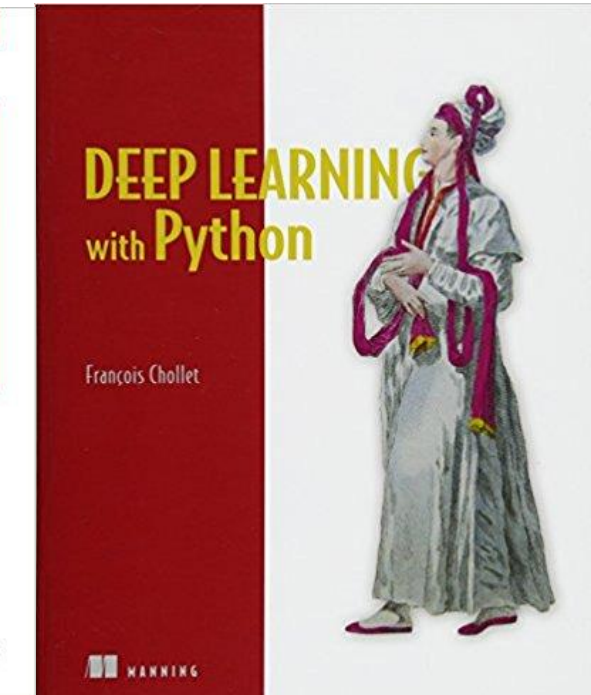






Applied Texts





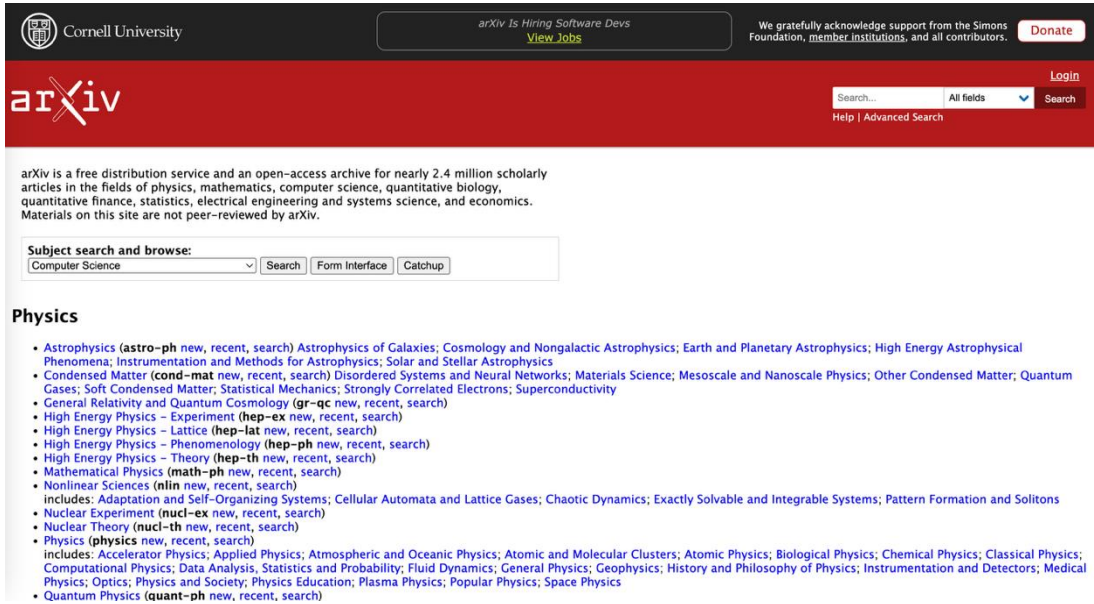
Staying up-to-date

Newsletters / Blogs / Channels:
Data Elixir (link)
LLMs - Ahead of AI, Sebastian Rashka (link)
Computer vision - Two Minute Papers, Károly Zsolnai-Fehér (link)

Corporate Blogs:
Google Research (link)
Microsoft Research (link)
DeepMind Research (link)
Kaggle (link)
Hugging Face (link)

Conferences:
ICML: International Conference on Machine Learning (link)
NeurIPS: Neural Information Processing Systems (link)
ICLR: International Conference on Learning Representations (link)
CVPR: IEEE Conference on Computer Vision and Pattern Recognition (link)
SIGKDD: ACM International Conference on Knowledge Discovery & Data Mining (link)

Arxiv (link)



Parting Thoughts

Parting advice

Most of machine learning is NOT modeling

Problem formulation, experimental design, and data quality are typically most important

Consider the consequences of your choices for others

Bias, fairness, privacy, and misuse are real concerns. This affects people.

Baby steps: start with a simple model, add complexity as needed

Understand your problem BEFORE you optimize anything

Be suspicious of all results (including your own!)

Verify experimental designs, data quality, and when results look too good to be true, they usually are

Be prepared to iterate

It's rare to get it right at the start, learn more about your problem, retry

Be a lifelong learner (and a humble one)

This field will continue to evolve quickly and will require continuous learning – not just ML, but probability, statistics, etc. Be humble and always open to learning.

Educating the **mind** without educating the **heart** is no education at all.

Aristotle